MUST S.1 **Self-containment** 1/3

All modules and components essential to replicate computational execution MUST be reachable within a single top-level research object.

- Are all files and directories nested under a common root? ✓ Yes ✗ No
- Are datasets included, linked, or referenced reachable from the code and environment specifications without crossing the boundary of a common root? ✓ Yes ✗ No
  - Reason for not meeting requirement...
- Is external software included in the environment or linked as submodules within the project boundary? ✓ Yes ✗ No

MUST T.1 + T.3 + T.5 **Tracking** 0/3

Persistent content identification MUST be recorded for all components. Provenance of all modifications MUST be recorded. Provenance records MUST include the versions of all components involved.

- Are version control systems such as `Git` used for code, text, documentation, and configuration files? ✓ Yes ✗ No
- Are version control systems such as `git-annex`, DataLad, or `Git` LFS used for large binary data? ✓ Yes ✗ No
- Are the exact environment specifications used to generate a set of results included in the provenance records to link computational actions to particular environments? ✓ Yes ✗ No

MUST A.1 **Actionability** 0/2

Research object MUST contain sufficient instructions to reproduce all computational results.

- Is a `README.md` or `Makefile` included with instructions for installation and usage? ✓ Yes ✗ No
- Is there a clear starting point for users to start reproducing results (e.g., a main script, a workflow definition, or a container image)? ✓ Yes ✗ No

MUST P.1 **Portability** 1/3

Procedures MUST NOT depend on undocumented host environment state.

- Are relative paths used in scripts, avoiding hardcoded or system-specific paths like `C:\Users\...` or `/home/user/...`? ✓ Yes ✗ No
- Are all software dependencies included in the environment specification, rather than relying on pre-installed tools? ✓ Yes ✗ No
- Have all assumptions about the host system's configuration been documented (e.g., specific OS versions, required system libraries, or environment variables)? ✓ Yes ✗ No

MUST P.2 **Portability** 0/1

Computational environments MUST be explicitly specified.

- Is there a clear list of system requirements and dependencies documented in the README or environment specifications? ✓ Yes ✗ No

MUST P.3 **Portability** 0/2

Environment definitions MUST be version controlled.

- Are environment specifications (e.g., Dockerfiles, `pyproject.toml`, `package.json`) included in the version control system alongside code and data? ✓ Yes ✗ No
- Is there a process for updating environment specifications when dependencies change, and are these updates tracked in version control? ✓ Yes ✗ No

MUST D.1 **Distributability** 1/3

All referenced modules and components MUST be persistently retrievable by others.

- Are environment specifications (e.g., container digests, frozen package manifests) included to ensure others can attempt to exactly replicate the environment? ✓ Yes ✗ No
- Are environment specifications shared in a way that others can access and use them (e.g., published container images, archived environment files)? ✓ Yes ✗ No
- Is there documentation on how to obtain and use the environment specifications for reproduction? ✓ Yes ✗ No

SHOULD T.2 **Tracking** 0/1

All components SHOULD be tracked using the same content-addressed version control system.

- Is a common version control system (e.g., `Git`, DVC, `git-annex`, or DataLad) used across all components? ✓ Yes ✗ No

SHOULD A.2 **Actionability** 0/1

Procedures SHOULD be specified as executable specifications.

- Is the workflow tested regularly to ensure instructions remain accurate? ✓ Yes ✗ No

SHOULD M.1 **Modularity** 0/1

Components SHOULD be organized in a modular structure.

- Are raw data, processed data, code, and environment definitions separated into distinct modules? ✓ Yes ✗ No

SHOULD E.1 **Ephemerality** 0/2

Computational results SHOULD be produced in ephemeral environments.

- Is the pipeline tested in a fresh container, batch job, clean virtual machine, or cloud-based instance? ✓ Yes ✗ No
- Does the workflow spawn disposable environments for each execution, allowing for isolation of runs? ✓ Yes ✗ No

SHOULD D.2 **Distributability** 0/3

Environment specifications SHOULD support reproducible builds.

- Are the exact environment specifications used to generate a specific set of results regularly tested to ensure they can be reconstituted as intended? ✓ Yes ✗ No
- Are environment artifacts archived within the research object where possible (e.g., executable binaries, container images)? ✓ Yes ✗ No
- Is there a process for updating environment specifications when dependencies change, and are these updates tracked in version control? ✓ Yes ✗ No

MAY M.2 **Modularity** 0/5

Components MAY be included directly or linked as subdatasets.

- Are external datasets or software dependencies included as submodules or linked as submodules? ✓ Yes ✗ No
- Is the modular structure documented to clarify how components relate and can be recombined? ✓ Yes ✗ No
- Are modular boundaries defined in a way that allows independent updates without breaking the overall workflow? ✓ Yes ✗ No
- Is there a clear mechanism for composing modules together (e.g., `git` submodules, or container orchestration)? ✓ Yes ✗ No
- Is there documentation or tooling to support users in understanding and navigating the modular structure? ✓ Yes ✗ No